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Anion Movement in a Soil under Pasture

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Abstract

Laboratory and field studies of anion movement in soils were made. In the laboratory the movement of a solution containing chloride and phosphate through saturated field cores (143 mm in diameter and 170 mm long) was studied. Both the anions appeared in the effluent almost immediately after application to the soil surface. Dye studies showed the movement was highly preferential, occurring mainly through biogenic macropores such as root and worm channels.

Unsaturated solute flow was also studied in the laboratory. When the matric potential in these field cores was reduced from zero to $-2\cdot0$ J/kg (-20 mb), the volumetric water content decreased from $0\cdot56$ to $0\cdot53$, while the hydraulic conductivity decreased two orders of magnitude. The flow of surface-applied bromide and dye in the slightly unsaturated soil was much less preferential than in the saturated soil, a finding that agrees in general terms with the predictions of theory previously published.

In the field, the flow of a surface-applied chloride and phosphate solution to a mole drain (at $0\cdot4$ m depth) was also highly preferential. In one experiment the concentration of both chloride and phosphate in the mole effluent exceeded $0\cdot8$ of that in the applied solution within 5 min of its application to the soil surface, when only 20 mm of solution had infiltrated. Dye studies again showed root and worm channels, sometimes in association with incipient fracture planes, were the preferential pathways. This indicates the importance of biological factors in maintaining hydraulic conductivity and mole drain performance.

Introduction

Much of the work done on solute movement in soil has involved the use of repacked soil. The resulting movement and dispersion can in this case usually be described adequately by the equation

$$\frac{\partial C}{\partial t} = D\frac{\partial^2 C}{\partial x^2} - v\frac{\partial C}{\partial x}, \tag{1}$$

which applies both for saturated and unsaturated flow. In this equation C is the solute concentration in the soil solution, t is time, D is the dispersion coefficient, x is distance, and v the mean pore velocity along the x-axis. The soil is thus treated as a homogeneous continuum. For non-sorbed ions the breakthrough curves observed from repacked soil columns with steady flow and a step-function change in concentration at the surface are usually sigmoid in shape, and approximately symmetrical about one pore volume (Biggar and Nielsen 1967). This indicates a fairly uniform displacement of the soil solution by the invading solution. The breakthrough curves for adsorbed ions are of similar shape, but considerably retarded.

Experiments carried out in the field, or in the laboratory using relatively undisturbed soil cores, have however often indicated much less-uniform displacement. Deeper penetration or earlier breakthrough than expected has been attributed to flow down preferential pathways (Wild and Babiker 1976; Anderson and Bouma

1977*a*, 1977*b*). Some of the consequences of such preferential flow have recently been reviewed by Thomas and Phillips (1979).

If the soil consists of large aggregates or contains continuous macropores, miscible displacement cannot be described by equation (1). The theory of dispersion in aggregated porous media is well developed, papers by van Genuchten and Wierenga (1976) and Rao *et al.* (1980*a*, 1980*b*) presenting recent work. A simple model of solute flow through a soil containing macropores was described by Scotter (1978), and we conducted experiments using moulded soil containing artificial macropores (Kanchanasut *et al.* 1978). This work indicated that, if the soil and flow conditions necessary for preferential flow occurred, it was of little import whether or not the solute involved was capable of being adsorbed by the soil matrix. We predicted that both non-sorbed and strongly adsorbed solutes would exhibit similar early breakthrough or deep penetration. Described in this paper are miscible displacement experiments aimed at confirming this prediction, using relatively large cores in the laboratory, and using the flow to mole drains in the field.

Scotter (1978) has suggested that preferential flow could only occur in pores larger than a certain diameter, in which the convective viscous flow of solute was greater than the molecular diffusion of the solute into the surrounding soil matrix. These larger pores would be drained at a matric potential of $-1\cdot5$ J/kg. Thus he suggested that relatively uniform miscible displacement would occur in unsaturated soil if the matric potential was lower than $-1\cdot5$ J/kg. Also described here is an experiment designed to check this prediction.

A problem with unsaturated miscible displacement experiments using soil cores is the unknown effect of the porous plates usually placed at either end of the cores (e.g. Elrick and French 1966). Dispersion in the plates probably masks, to some extent at least, any preferential flow occurring in the soil. Poor contact and pore blockage at the core–plate interface can also be a problem. The experimental techniques described in this paper minimized these effects, while maintaining a uniform matric potential throughout the core.

Materials and Methods

Soil Core Experiments

Miscible displacement of anions was investigated using 'undisturbed' soil cores under both saturated and unsaturated flow conditions. The apparatus used is shown in Fig. 1. Cylindrical cores were obtained during winter from the top 150 mm of Tokomaru silt loam (a Fragiaqualf) at the site described by Scotter *et al.* (1979). Aluminium corers of 143 mm inside diameter and 170 mm long, with an area ratio of 0·07 (Loveday 1974) were used. The cores were taken when the soil was at or near field capacity. They were carefully excavated by removing the soil around each corer to slightly deeper than the core depth and then breaking off the core. The excess soil was removed from the base by gently chipping off aggregates along the natural fracture planes, to prevent smearing and sealing of conducting pores. Nylon mesh with an effective pore diameter of 60 μm was then sealed to the base of the corers. This mesh has a sufficiently high air-entry value, but a much higher permeability and lower pore volume than porous ceramic. Tensiometers were installed, as indicated in Fig. 1.

The soil columns were initially saturated from below with a solution containing 0·001 M potassium nitrate and 0·0015 M sodium azide, and were then leached with the same solution. A head of approximately 10 mm was maintained on the soil surface. The sodium azide served as a bacterial inhibiter. When the flux was constant and the tensiometers indicated an hydraulic gradient approaching unity, application of this solution ceased. When the ponded solution had just disappeared from the soil surface, it was replaced by a new solution containing 0·003 M potassium chloride, $6 \cdot 5 \times 10^{-4}$ M phosphorus as potassium dihydrogen phosphate and 0·0015 M sodium azide. The potassium nitrate concentration in the displaced solution was chosen so that the two solutions had the same density, to avoid gravity segregation effects.

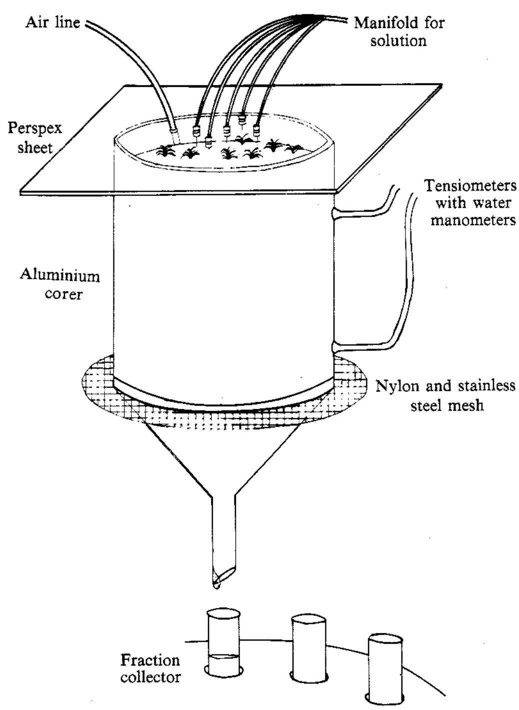

Fig. 1. Apparatus for soil core experiments.

The effluent aliquots were analysed for chloride using a specific ion electrode and for phosphorus as described by Murphy and Riley (1962). After approximately 2 pore volumes of effluent had been obtained, 2 pore volumes of methylene blue dye solution (0·001 g/g aqueous solution) were applied to each column. The saturated cores were weighed at the end of this experiment to allow the pore volume to be determined.

Next the soil cores were desaturated slightly by applying a positive air pressure of 0·02 bar to the sealed chamber above the soil surface and reducing the inflow rate by small increments. This was done until the two tensiometers indicated a matric potential of $-2 \cdot 0 \pm 0 \cdot 1$ J/kg ($-0 \cdot 02$ bar) and steady-state conditions again prevailed. The influent solution was then changed to a solution containing 0·0063 M potassium bromide, 0·0015 M sodium azide and 1% by weight of rhodamine B dye. This solution was introduced onto the soil cores through a manifold with five small outlets, distributing the solution fairly uniformly over the soil surface. Effluent aliquots were collected and analysed for bromide using a specific ion electrode. No rhodamine was visible in any of the aliquots.

At the conclusion of the experiment, the cores were again weighed, and then oven-dried to constant mass at 105°C. The saturated and unsaturated liquid-filled pore volumes and water contents were then calculated.

Mole Drain Experiment

The field experiment was conducted at a site adjacent to where the soil cores were taken. A mole-tile drainage system had been installed 6 years earlier. Mole drains 60 mm in diameter were pulled at 2 m spacing in the clay loam B horizon at approximately 400 mm depth.

The experiment was carried out in late autumn and repeated in late winter; at both times the soil was at 'field capacity', but the drains were not running. A thin-walled, galvanized iron infiltrometer ring, 380 mm in diameter, was driven in to 100 mm depth above a mole line. The soil surrounding the infiltrometer ring was wetted, and water was ponded in the ring until the infiltration rate and mole drain flow were nearly equal. The ponded water was then replaced with a solution containing 0·003 M of potassium chloride and $1\cdot3\times10^{-3}$ M phosphorus as potassium dihydrogen phosphate. A head of 20–25 mm was maintained in the ring. Aliquots were collected from the drain flow and analysed for chloride and phosphate as described above. At the end of the experiment 25 l. of methylene blue dye solution were infiltrated through the ring, the soil was allowed to drain for 2 days, then the profile was exposed and the dye movement pathways observed.

Results and Discussion

Saturated Flow Through Cores

Data relating to the core experiments are given in Table 1, and the breakthrough data are presented in Fig. 2. The results for the saturated cores show that both

Table 1. Physical data for core experiments

	Hydraulic conductivity (m s^{-1})	Volumetric water content	Time for one pore volume (h)	Dispersion coefficient (m^2 s^{-1})
Core I				
Saturated	$2\cdot4\times10^{-5}$	0·562	0·90	–
Unsaturated[A]	$1\cdot0\times10^{-7}$	0·535	215	$1\cdot8\times10^{-8}$
Core II				
Saturated	$1\cdot7\times10^{-5}$	0·558	1·3	–
Unsaturated[A]	$1\cdot6\times10^{-7}$	0·531	137	$5\cdot8\times10^{-9}$

[A] At a matric potential of $-2\cdot0$ J/kg.

chloride and phosphorus appeared in the effluent almost immediately after application to the soil surface, indicating highly preferential flow. This is in sharp contrast to the breakthrough curves resulting from other experiments, not described in detail here, using packed columns of 0·5–1 mm aggregates of the same soil. With the same applied anion concentrations, and a similar flux density of $4\cdot7\times10^{-5}$ m/s, the relative concentration (ratio of effluent to influent concentration) of chloride and phosphate did not reach 0·5 until 0·94 and 8·3 pore volumes respectively had passed through the soil (Kanchanasut 1980).

The breakthrough data for the saturated cores cannot be described by the appropriate solution to equation (1), so it cannot be used to calculate a dispersion coefficient. The differences between the chloride and phosphate data indicate adsorption has had an observable effect, particularly in Core II. However, as indicated by Kanchanasut *et al.* (1978), this can probably be largely attributed to diffusion and adsorption of phosphate directly from the ponded solution before it enters the soil, rather than adsorption during viscous flow through the soil.

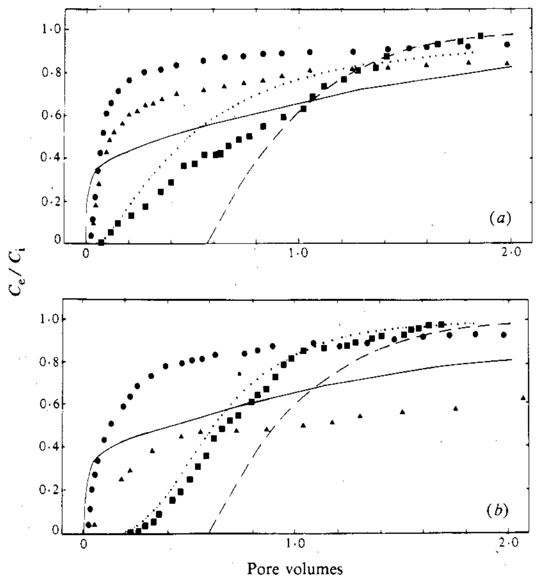

Fig. 2. Breakthrough data from soil cores for chloride (●) and phosphate (▲) during saturated flow, and for bromide (■) during unsaturated flow at −2 J/kg matric potential. Also shown are computed breakthrough curves for bromide assuming the flow occurs in channels 0·15 mm (——) and 0·10 mm (– – –) in diameter after Scotter (1978), and curves found by fitting the solution of equation (1) to the bromide data (······). C_e/C_i is the ratio of effluent to influent concentration. (*a*) Core I. (*b*) Core II.

In Fig. 3 is shown a cross-section of one of the cores, indicating the areas dye-stained during saturated flow. Only a small fraction of the soil was stained by dye. Breaking the soil cores along natural fracture planes showed some staining there, as well as of root and worm channels. Often the dye had moved through a system of interconnected macropores of different origin. Worm channels did not necessarily have to reach the surface to be effective, provided they were connected to other somewhat smaller channels or cracks that did.

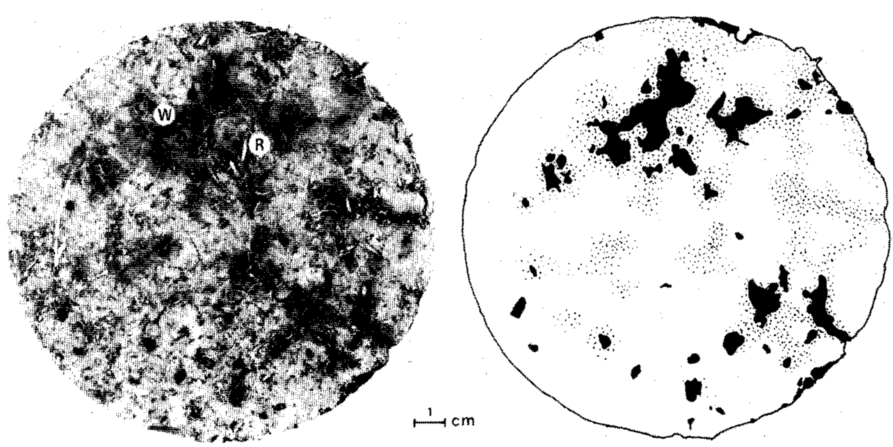

Fig. 3. (*a*) A cross-section of core II, cut 20 mm below the soil surface, showing macropores (W, worm channel; R, root channel). (*b*) Diagram showing areas dye-stained during saturated flow (shaded black) and areas stained during unsaturated flow (speckled), as indicated on a colour print of (*a*).

Unsaturated Flow Through Cores

It is interesting to note from Table 1 that, while the small change in matric potential from 0 to $-2 \cdot 0$ J/kg induced a change of only $0 \cdot 03$ in volumetric water content, it brought about a two orders of magnitude decrease in hydraulic conductivity. Germann and Beven (1981), also using undisturbed soil, obtained similar data.

The unsaturated breakthrough data for bromide are shown in Fig. 2. At a pressure potential of $-2 \cdot 0$ J/kg preferential flow would not be expected, according to the model of Scotter (1978), and in fact the displacement is a lot more uniform than when the columns were saturated. The relative concentration did reach $0 \cdot 5$ before one pore volume, however, suggesting the displacement was not entirely uniform. There are a number of possible reasons for this which are discussed below.

Firstly, the model assumed that the flow in the soil was through uniformly spaced, cylindrical channels all of the same diameter. In real soils macropores are neither uniform cylinders, nor are they regularly spaced. They are likely to have constrictions, and to be randomly spaced, or even tending to be clumped together. A short constriction at the top of a larger pore will control the matric potential at which the pore drains, but such a pore would conduct much more solution than a uniform pore with the diameter of the constriction. For example, solution of the Hagen–Poiseuille equation for a 1 mm diameter cylindrical channel with a 2 mm long constriction $0 \cdot 15$ mm in diameter at the top shows it would conduct 75 times as much solution as a uniform $0 \cdot 15$ mm diameter channel, although both would drain at a matric potential of $-2 \cdot 0$ J/kg. Thus some preferential flow could still occur at a lower matric potential than the simple theory predicts.

Secondly, random or clumped arrangements of conducting pores will lead to somewhat earlier breakthrough than regularly spaced conducting pores, as the diffusion zones around adjacent viscous flow pathways will overlap each other more quickly in some places. However, the breakthrough from other parts of the soil where the conducting pores are spaced more sparsely will be delayed, causing 'tailing' of the breakthrough curves. Other possible reasons for the early breakthrough of the bromide are the method of application of solution to the surface of the cores, using five point sources rather than uniform application over the whole surface, and anion exclusion. The relatively low exchange capacity of this soil (Pollok 1975), and the results of miscible displacement experiments using sieved aggregates of the same soil (Kanchanasut 1980), however, suggest that anion exclusion was not significant.

Also shown in Fig. 2 are the curves obtained by fitting a solution of equation (1) for the appropriate boundary conditions to the bromide data, using the technique described by Rose and Passioura (1971). The agreement is only fair; the dispersion coefficients found are listed in Table 1. Assuming a tortuosity factor of $0 \cdot 4$, and a diffusion coefficient for dilute potassium bromide solution of 2×10^{-9} m^2/s (Robinson and Stokes 1959), the dispersion coefficients are an order of magnitude greater than the molecular diffusion coefficient in the soil.

The other theoretical breakthrough curves in Fig. 2 were computed using the simple theory of Scotter (1978), assuming that all viscous flow occurred through uniformly spaced, cylindrical channels $0 \cdot 15$ or $0 \cdot 10$ mm in diameter. The spacing between the channels calculated to give the measured unsaturated hydraulic conductivities ranged from 7 to 20 mm. The very different curve shapes for the two assumed pore diameters indicates how sensitive the breakthrough curves are to pore size over this range.

The staining by rhodamine dye during unsaturated flow is shown by the speckled area in Fig. 3. The pattern shows non-uniform flow, although the breakthrough data indicate the non-sorbed bromide moved into a much larger soil volume than the dye. This was presumably due to the dye being adsorbed by the soil. There was a tendency for the rhodamine-stained soil to surround the methylene blue-stained soil. This suggests some pathways were used for both saturated and unsaturated flow, perhaps partly due to surface constrictions as discussed above. It was also observed that smaller and larger macropores often occurred in clumps, perhaps due to interactions between the causal biological factors, or to both roots and worms exploiting the same more easily penetrated regions of the soil.

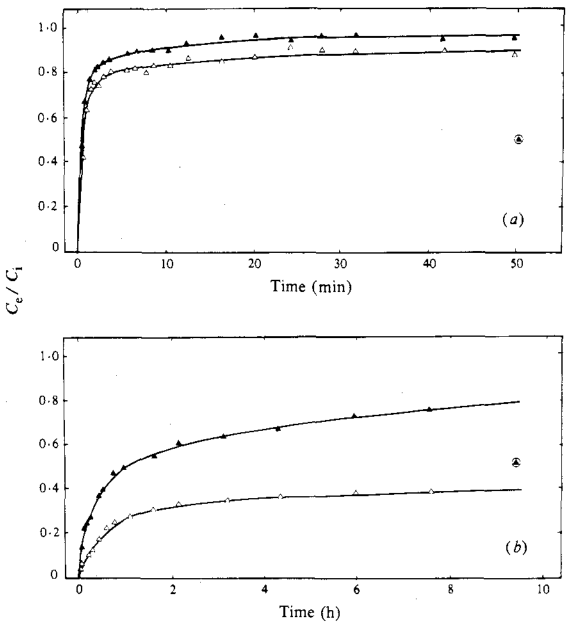

Fig. 4. Breakthrough data for the mole drain; chloride (▲) and phosphorus (△). The circled symbols indicate approximate times for one pore volume of flow. C_e/C_i is the ratio of effluent to influent concentration. (*a*) Late autumn experiment. (*b*) Late spring experiment. The lines have been fitted by eye.

Flow to a Mole Drain

As there was free water on the soil surface and in the drain, the flow in this experiment was essentially under saturated conditions. Breakthrough data, showing the chloride and phosphate concentrations in the mole drain effluent, are presented in Fig. 4. Both anions appeared in drain effluent very soon after application to the soil surface, indicating highly preferential flow. In the first experiment (Fig. 4*a*), which was conducted in late autumn, the steady-state infiltration rate was $6 \cdot 8 \times 10^{-5}$ m/s. The relative concentration of both ions in the mole effluent rose quickly to more than 0·8 within 5 min, when only 20 mm or approximately 0·1 pore volumes of solution had entered the soil (regarding the pore volume as being the porosity times the volume of a cylinder with diameter equal to that of the infiltrometer ring and extending to mole depth).

When the experiment was repeated at an adjacent site in late winter (Fig. 4*b*) the infiltration rate was $6 \cdot 7 \times 10^{-6}$ m/s, an order of magnitude less than in late autumn. The reason for this is suggested later; however, preferential flow still occurred. The relative concentration reached 0·5 for chloride and 0·25 for phosphate after 1 h, when 24 mm of solution had infiltrated.

The staining pattern of the methylene blue applied to the soil after the winter experiment showed that the dye solution moved mainly through large pores, such as the gaps between roots and soil, the channels left by decaying roots, worm channels, and the incipient fracture planes between structural units. Commonly some combination of these pore types provided a continuous pathway from the surface to the mole. Grass root channels and worm channels were the main pathways for water entry into the soil profile. Some dye-stained worm channels extended from the surface nearly to mole depth. Grass roots ramified through the soil to beyond mole depth, with some roots penetrating the mole channel itself and providing a pathway for flow into it. Other grass roots were observed to have followed old worm channels. The dye staining tended to be concentrated in the soil directly above the mole, which had been disturbed 6 years earlier when the mole was pulled.

Staining of incipient fracture planes was also commonly observed, with plant roots and worm channels also tending to follow the fracture planes, particularly in the B horizon. These fracture planes were probably, to some extent at least, induced or enhanced by moling. Bowler (1980) states that moling causes the soil to become fractured and permeated by cracks. He comments that these cracks tend to close when the soil is wet, but reopen when the soil begins to dry. This would explain the faster and more preferential flow in late autumn compared to late winter. In late autumn the soil had only recently rewet, and the closing of the cracks is probably quite a gradual process in the B horizon, due to the low hydraulic conductivity of the soil between preferential flow pathways.

Conclusions

Both laboratory and field studies showed saturated flow in this soil, which had been under pasture for a long period, was highly preferential. Root and worm channels, sometimes in association with incipient fracture planes, were the main preferential pathways. If these pathways had been absent the saturated hydraulic conductivity would have been much lower. It would appear that macrobiological activity in the soil is vital for maintaining the long-term effectiveness of mole drains.

The unsaturated breakthrough data from the cores were much less preferential, tending to confirm the prediction made by Scotter (1978) that at a matric potential of $-2\cdot0$ J/kg the flow would be relatively uniform. At this matric potential most soils are still considerably wetter than 'field capacity'.

While surface ponding during heavy rainfall has been observed at the site, it is fairly rare, as can be inferred from the measured steady infiltration rates of 240 and 24 mm/h. However, the results of the field experiment indicate what could happen if dairy shed or factory waste is applied to the soil at a rate to cause surface ponding. Virtually untreated waste could appear in the drainage system within minutes of its application to the surface. In fact at an adjacent site A. N. Macgregor (as reported by Kanchanasut *et al.* (1978)) found urea and coliforms in the mole-tile drainage system within 2 h of spraying dairy shed waste on the soil at 10 mm/h in early spring.

Conversely, nutrients, such as nitrate, present in the soil solution will be relatively less prone to leaching when preferential rather than uniform flow dominates.

The observed preferential movement of phosphate in the field shows that even strongly adsorbed ions or molecules can sometimes move from the soil surface to mole drains at 400 mm depth within minutes. Thus heavy rain soon after superphosphate spreading could lead to significant phosphate concentrations in the drainage water. Even 10 mm of rain can have a measurable effect, as P. E. H. Gregg (reported by Kanchanasut *et al.* (1978)) observed at the same site. He found 10 mm of rain on the night following superphosphate application resulted in a 10-fold increase (from $0\cdot1$ to $1\cdot0$ μg/ml) over the preceding drainage event in the dissolved inorganic phosphate concentration in the drain effluent.

References

Anderson, J. L., and **Bouma, J.** (1977*a*). Water movement through pedal soils. I. Saturated Flow. *Soil Sci. Soc. Am. J.* **41**, 413–18.

Anderson, J. L., and **Bouma, J.** (1977*b*). Water movement through pedal soils. II. Unsaturated flow. *Soil Sci. Soc. Am. J.* **41**, 419–23.

Biggar, J. W., and **Nielsen, D. R.** (1967). Miscible displacement and leaching phenomenon. *In* 'Irrigation of Agricultural Lands', *Agronomy* **11**, 254–74. (Eds. R. M. Hagan, H. R. Haise and T. W. Edminster.) (Am. Soc. Agron.: Madison, Wisc.).

Bowler, D. G. (1980). 'The Drainage of Wet Soils.' (Hodder and Stoughton: Auckland.)

Elrick, D. E., and **French, L. K.** (1966). Miscible displacement patterns on disturbed and undisturbed soil cores. *Soil Sci. Soc. Am. Proc.* **30**, 153–6.

van Genuchten, M. Th., and **Wierenga, P. J.** (1976). Mass transfer studies in sorbing porous media. I. Analytical solutions. *Soil Sci. Soc. Am. J.* **40**, 473–80.

Germann, P., and **Beven, K.** (1981). Water flow in soil macropores. I. An experimental approach. *J. Soil Sci.* **32**, 1–13.

Kanchanasut, P. (1980). Anion movement in a structured soil. Ph.D. Thesis, Massey University, Palmerston North.

Kanchanasut, P., **Scotter, D. R.**, and **Tillman, R. W.** (1978). Preferential solute movement through larger soil voids. II. Experiments with saturated soil. *Aust. J. Soil Res.* **16**, 269–76.

Loveday, J. (1974). Methods for soil analysis of irrigated soils. Tech. Commun. No. 54. (Commonwealth Agricultural Bureaux: Farnham Royal, U.K.).

Murphy, J., and **Riley, J. P.** (1962). A modified single solution method for the determination of phosphate in natural water. *Anal. Chim. Acta* **27**, 31–6.

Pollok, J. A. (1975). A comparative study of certain New Zealand and German soils formed from loess. Ph.D. Thesis, Friedrich-Wilhelms-Universität, Bonn.

Rao, P. S. C., **Jessup, R. E.**, **Rolston, D. E.**, **Davidson, J. M.**, and **Kilcrease, D. P.** (1980*a*). Experimental and mathematical description of non-adsorbed solute transfer by diffusion in spherical aggregates. *Soil Sci. Soc. Am. J.* **44**, 684–8.

Rao, P. S. C., **Rolston, D. E.**, **Jessup, R. E.**, and **Davidson, J. M.** (1980*b*). Solute transport in aggregated porous media: theoretical and experimental evaluation. *Soil Sci. Soc. Am. J.* **44**, 1139–46.

Robinson, R. A., and **Stokes, R. H.** (1959). 'Electrolyte Solutions.' 2nd Edn. (Butterworths: London.)

Rose, D. A., and **Passioura, J. B.** (1971). The analysis of experiments on hydrodynamic dispersion. *Soil Sci.* **111**, 252–7.

Scotter, D. R. (1978). Preferential solute movement through larger soil voids. I. Some computations using simple theory. *Aust. J. Soil Res.* **16**, 257–67.

Scotter, D. R., **Clothier, B. E.**, and **Corker, R. B.** (1979). Soil water in a Fragiaqualf. *Aust. J. Soil Res.* **17**, 443–53.

Thomas, G. W., and **Phillips, R. E.** (1979). Consequences of water movement in macropores. *J. Environ. Qual.* **8**, 149–52.

Wild, A., and **Babiker, I. A.** (1976). The asymmetric leaching pattern of nitrate and chloride in a loamy sand under field conditions. *J. Soil Sci.* **27**, 460–6.

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